

Green Guide : Plant Report

A tomato plant in a home garden exhibited signs of distress including yellowing leaves, black spots, and curled edges, the plant was diagnosed with Early Blight – a common fungal disease caused by *Alternaria solani*. This disease typically starts on older leaves and progresses upward, leading to premature leaf drop, reduced photosynthesis, and lower fruit yield. Early Blight thrives in warm, humid conditions and often affects tomato and potato plants. If not treated promptly, the infection can spread rapidly and compromise the entire crop.

Classification:

- Plant: Tomato (*Solanum lycopersicum*)
 - Disease: Early Blight
 - Pathogen Type: Fungal
 - Causal Agent: *Alternaria solani*
 - Infection Target: Primarily leaves, stems, and sometimes fruit
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Remedies

To manage Early Blight, a combination of cultural, organic, and chemical methods can be used. It is important to improve airflow, avoid overhead watering, and rotate crops. Organic remedies such as neem oil or compost tea can suppress fungal growth in early stages.

1. Cultural Control:

- Rotate crops annually to avoid soil-borne spores.
- Remove and destroy infected plant debris.
- Space plants properly to enhance airflow.

2. Watering Tips:

- Water at the base to keep leaves dry.
- Avoid overhead irrigation, especially in the evening.

3. Organic Solutions:

- Spray neem oil weekly as a preventive measure.
- Apply compost tea to improve leaf resistance.

4. Chemical Treatment:

- Use copper-based fungicides for moderate infection.
- Apply chlorothalonil-based sprays under professional guidance.

5. Monitoring:

- Regularly inspect leaves for early signs.
- Remove affected leaves immediately to contain spread.